



B.Tech. Degree VII Semester Special Supplementary Examination April 2025

ME 19-205-0701 REFRIGERATION AND AIR-CONDITIONING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of different refrigeration cycles.
 CO2: Study the vapour compression and vapour absorption refrigeration cycles- both theoretical and practical cycles, properties of refrigerants, and selection criteria for refrigerants.
 CO3: Understand the different components of refrigeration systems.
 CO4: Get knowledge of psychrometry, air-conditioning systems and will be able to do the cooling load calculation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

(Refrigeration data books are permitted for examination)

PART A(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a) Explain the difference between the terms thermal efficiency and coefficient of performance.	3	L2	1	1,2,4	
	(b) A heat engine having an efficiency of 70% is used to drive a refrigerator having a coefficient of performance of 5. Calculate energy absorbed from low temperature reservoir by refrigerator for each kJ of energy absorbed from high temperature source by engine.	3	L3	1	1,2,4	
	(c) How actual Vapor Compression Refrigeration System is different from ideal Vapor Compression Refrigeration System?	3	L1	2	1,2,4	
	(d) In Vapor Absorption Refrigeration System, why is the fluid in the absorber cooled and fluid in generator heated.	3	L2	2	1,2,4	
	(e) Write short notes on flooded evaporators.	3	L1	3	1,2,4	
	(f) Explain the working principle of reciprocating compressors with neat figures.	3	L2	3	1,2,4	
	(g) Explain following processes with Psychrometric chart in detail (i) Cooling and dehumidification (ii) Heating and dehumidification.	3	L2	4	1,2,4	
	(h) Write short notes on human comfort.	3	L2	4	1,2,4	

PART B

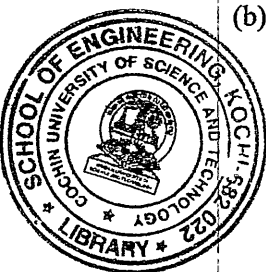
(4 × 12 = 48)

- II. (a) Explain thermo-electric refrigeration with a neat figure. 6 L1 1 1,2,4
 (b) In gas refrigeration system working on Joule cycle, the outlet temperature from the cold space is 270 K and temperature at inlet to turbine is 318 K. The pressure ratio is 4. Determine the mass flow rate, heat rejection, compressor work, turbine work, COP and volume flow rates of air at inlet to compressor and at outlet of turbine for a system of 1 TR cooling capacity. Working substance is air. 6 L3 1 1,2,4

OR

- III. (a) Explain the construction and working of vortex tube refrigeration system with a neat diagram. 6 L2 1 1,2,4
 (b) Write short notes on Bell-Coleman cycle and derive an equation for COP. 6 L2 1 1,2,4

(P.T.O.)



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		Marks	BL	CO	PO
IV.	(a) Write short notes on (i) Environment friendly refrigerants and its properties. (ii) Selection criteria of refrigerants.	6	L2	2	1,2,4
	(b) Explain multistage refrigeration systems with neat figures. What are the advantages of multistage systems over single stage?	6	L2	2	1,2,4
OR					
V.	(a) Explain aqua-ammonia VARS system with a neat figure.	6	L2	2	1,2,4
	(b) Refrigerant-134a enters compressor as superheated vapour at 0.14 MPa and -10°C at a rate of 0.05 kg/s and leaves at 0.8 MPa and 50°C. The refrigerant is cooled in condenser to 26°C and 0.72 MPa and is throttled to 0.15 MPa. Disregarding any heat transfer and pressure drops in the connecting lines between the components, determine (i) the rate of heat removal from the refrigerated space and the power input to the compressor (ii) the isentropic efficiency of the compressor (iii) the coefficient of performance of the refrigerator.	6	L3	2	1,2,4
VI.	(a) Explain different types of condensers with neat figures.	6	L2	3	1,2,4
	(b) Explain how single-stage and multi-stage compressors differ in their working principles and applications. Include clear, labeled diagrams to support your explanation.	6	L1	3	1,2,4
OR					
VII.	(a) Explain the working principles of screw and vane type compressors, using clear, labeled diagrams to illustrate their operation.	6	L2	3	1,2,4
	(b) Write short notes on different types of expansion devices with neat figures.	6	L1	3	1,2,4
VIII.	(a) Explain following terms with neat figures. (i) Air distribution systems. (ii) Air handling systems.	6	L1	4	1,2,4
	(b) An air-conditioning system is to take in outdoor air at 10°C and 30 percent relative humidity at a steady rate of 45 m ³ /min and to condition it to 25°C and 60 percent relative humidity. The outdoor air is first heated to 22°C in the heating section and then humidified by the injection of hot steam in the humidifying section. Assuming the entire process takes place at a pressure of 100 kPa, determine (i) the rate of heat supply in the heating section. (ii) the mass flow rate of the steam required in the humidifying section.	6	L3	4	1,2,4
OR					
IX.	(a) Write short notes on central and unitary air-conditioning systems.	6	L2	4	1,2,4
	(b) Two airstreams are mixed steadily and adiabatically. The first stream enters at 35°C and 30 percent relative humidity at a rate of 15 m ³ /min, while the second stream enters at 12°C and 90 percent relative humidity at a rate of 25 m ³ /min. Assuming that the mixing process occurs at a pressure of 1 atm, determine the specific humidity, the relative humidity, the dry-bulb temperature and the volume flow rate of the mixture.	6	L1	4	1,2,4

Blooms's Taxonomy Levels

L1 – 30%, L2 – 52.5%, L3 – 17.5%.
